

Event Booking

Weekly Mini Project 02 — Team 4 · Project Specification

Project	Event Booking
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Find an event, reserve your place, and trust that every seat count is always right.

1. Project Overview

Workshops, talks, concerts and meetups have a limited number of places. When registration is handled by hand, events get overbooked, people register twice, and nobody knows how many places are really left.

Event Booking lets people browse upcoming events, see exactly how many places remain, reserve one or more places, and later find or cancel their booking. The system guarantees that an event is never booked beyond its capacity.

2. Team

The team leader and members are listed in the table at the top of this document.

Team Responsibility

- The leader coordinates the internal work split and makes sure the team reaches a working integrated result.
- Frontend/backend ownership is not assigned by the instructor.
- The team should decide how responsibilities are distributed.
- Every member must make a meaningful contribution.
- Every member should understand the project at a high level.
- Every member must participate in the final English presentation.

3. Project Goal

Build a web application where a visitor can:

- browse and filter upcoming events;
- see each event's details and how many places remain;
- book one or more places for an event;
- receive a booking confirmation with a booking code;
- later find their booking and cancel it, returning the places to the event.

Capacity, availability and booking decisions must always reflect the current bookings and remain consistent on every screen.

4. Technology Constraints

Area	Requirement
Frontend	React + TypeScript
Backend	NestJS

Area	Requirement
Data	In-memory only
Database	Not required

- The application's working data is kept in memory and may reset when the application server restarts.
- How you organize your application is your team's decision.
- Do **not** add a database, user accounts/login, external AI services, cloud services or paid APIs.

5. User Scenario

Sara wants to attend a design workshop next week. She opens Event Booking, filters by the category "Workshops", and finds "UI Design Basics" with a capacity of 30 and 27 places booked — **3 remaining, 90% booked**.

She books 2 places for herself and a friend, entering her name and phone number. The confirmation shows booking code **EV-2093**, and the event now shows **1 remaining**. When she accidentally submits the same booking again, it is rejected: she already has an active booking for this event.

A few days later her friend cannot come. Sara finds her booking with her phone number and code and cancels it. The event goes back to **3 remaining**, and another visitor who saw "Full" earlier can now book.

6. Required Features

Feature	Description
Browse events	View all upcoming events with their key information.
Filter events	Filter by category and date, and show only events with available places. Filters can be combined.
Event details	Open one event and see its full information and availability.
Available places	See how many places remain for every event, always up to date.
Book places	Book between 1 and 4 places for an event in one booking.
Booking confirmation	Receive a confirmation with a booking code and all booking details.
Find a booking	Find an existing booking using phone number + booking code.
Cancel a booking	Cancel an active booking, returning its places to the event.
Event availability summary	See each event's capacity, booked, remaining and booking percentage.

Event information

Each event must present at least:

- title
- description
- date
- time
- location
- category (for example: Workshop, Talk, Concert, Sports, Meetup)
- capacity
- number of booked places

Booking information

Each booking must present at least: booking code, event, customer name, customer phone number, number of places, booking date and status (**Active** or **Cancelled**).

7. Business Rules

Capacity

1. An event can **never** accept bookings beyond its capacity.
2. A booking is **rejected** when the event is full.
3. A booking is also rejected when it asks for more places than remain (for example, asking for 3 when 2 remain). Partial bookings are not created.
4. One booking can reserve between **1 and 4** places.
5. Available places must always reflect the current active bookings — after every booking and every cancellation.

Time

1. Past events cannot be booked. An event that has already started is treated as past.
2. Past events are not shown in the default event list. (They may be shown separately, clearly marked as past.)

Identity and duplicates

1. A customer name and a valid phone number are required to book.
2. One person (identified by phone number) cannot hold **two active bookings for the same event**. A second attempt is rejected with a clear message.
3. The same person can book different events.
4. Every booking receives a **unique booking code**, shown on the confirmation.
5. A booking can only be found when **both** the phone number and the booking code match.

Cancellation

1. Only an **Active** booking can be cancelled. A cancelled booking cannot be cancelled again.
2. A booking cannot be cancelled once the event has started.
3. Cancelling a booking makes all of its places available again immediately.
4. After cancelling, the same person may book the same event again (if places remain).

Event status

1. Every event shows one of: **Available**, **Almost full** (90% or more booked), **Full**, or **Past**.

8. Required Screens / Views

Screen	Must show / allow
Events	All upcoming events with title, category, date, time, location, remaining places and status; the filters. A clear message when no events match.
Event Details	All event information; the availability summary (capacity, booked, remaining, booking percentage, status); a way to book when booking is allowed.
Booking Form	The selected event and its remaining places; fields for name, phone number and number of places.
Booking Confirmation	Booking code, event title, date, time, location, number of places, status; the event's updated remaining places.
Find / Manage Booking	A form for phone number + booking code; the found booking's details and status; a cancel action that is only offered when cancellation is allowed.

All screens must be reachable through the application's navigation.

9. Required User Flows

Flow A — Discover events

1. The user opens the Events screen.
2. The user filters by category and/or date, or chooses to show only events with available places.
3. The list shows only matching events, each with its remaining places and status.
4. The user clears the filters and all upcoming events return.

Flow B — Book places

1. The user selects an event.
2. The application shows event details and remaining places.
3. The user enters name, phone number and number of places.
4. The booking is accepted or rejected according to the project rules.
5. If accepted, the user sees the Booking Confirmation, and the event's remaining places are reduced by the number booked.

Flow C — Event fills up

1. An event has only a few places left.
2. Bookings are made until no places remain.
3. The event now shows **Full** and 0 remaining.
4. Any further booking attempt for this event is rejected with a clear message.

Flow D — Find and cancel a booking

1. The user opens Find / Manage Booking.
2. The user enters phone number and booking code.
3. If both match, the booking details and status are shown; otherwise a clear "booking not found" message appears.
4. The user cancels the booking.
5. The booking becomes **Cancelled**, and the event's remaining places increase again.

10. Required Calculations / Derived Information

The application must produce and display:

Derived information	Where it appears
Capacity	Event Details
Number of booked places (from active bookings only)	Event Details
Number of remaining places	Events, Event Details, Booking Form, Confirmation
Booking percentage (booked ÷ capacity, as a percentage)	Event Details
Event status (Available / Almost full / Full / Past)	Events, Event Details
Booking status (Active / Cancelled)	Confirmation, Find / Manage Booking
Whether the booking can still be cancelled	Find / Manage Booking

These values must always agree with each other — for example, booked + remaining must always equal capacity.

11. Validation & Error Scenarios

General expectations

- Required information cannot be missing.
- Numeric values must be sensible (no negative, zero or absurd values where they make no sense).
- Invalid actions must be rejected — the system must never silently accept an invalid operation.
- The user must always receive a clear, human-readable message explaining **why** an action failed.
- Invalid operations must be rejected even if the user reaches them through an unexpected interaction (for example, an outdated page or a repeated submission).
- When a request cannot be completed or something goes wrong, the screen must show a sensible message — never a blank page or a frozen screen.

Project-specific scenarios

Situation	Expected behavior
Booking a full event	Rejected; the user is told the event is full.
Asking for more places than remain	Rejected; the user is told how many places remain.
Number of places is 0, negative, more than 4, or not a whole number	Rejected with a clear message.
Booking a past event	Rejected with a clear message.
Booking an event that does not exist	A clear "event not found" message.
Name or phone number missing, or phone number not valid	Rejected with a clear message.
The same phone number books the same event while an active booking exists	Rejected; the user is told they already have a booking for this event.
Phone number and booking code do not match any booking	A clear "booking not found" message.
Cancelling an already-cancelled booking, or after the event started	Rejected with a clear message.

12. In-Memory Data Requirements

No database is required.

- The application may start with seeded (pre-loaded) data.
- Data created or changed while the application is running must behave normally during that run — every screen must reflect the latest data, including after a page refresh.
- It is acceptable that all data resets when the application server restarts. This is an expected limitation of this assignment — do not spend time adding any other persistence technology.

For this project specifically:

- The application must start with at least **10 events** across at least **4 categories**, with different dates and capacities.
- The seeded data must include at least one **Full** event, at least one **Almost full** event, at least one **Past** event, and existing bookings — so every rule can be demonstrated.
- The booked number of each seeded event must match its seeded bookings.

13. Out of Scope

The following are **not required** and should not be built as part of the core project: user accounts, login or registration; online payment or ticket prices; QR code scanning; email or SMS delivery; a database; an administrator area for creating events.

14. Minimum Deliverable

Your project counts as complete only when **all** of the following work:

- ☐ A working React + TypeScript frontend.
- ☐ A working NestJS backend.
- ☐ The frontend and backend work together as one integrated application; every change a user makes is visible on all related screens.
- ☐ Every required screen in section 8 exists and is reachable through the application's navigation.
- ☐ Every business rule in section 7 is enforced.
- ☐ Invalid operations are rejected with useful, clear feedback.
- ☐ In-memory data changes correctly while the application is running.
- ☐ The project can be demonstrated from start to finish without manual fixes during the demo.
- ☐ Events can be browsed, filtered (combined) and opened in detail.
- ☐ Bookings of 1–4 places work and reduce the remaining places.
- ☐ Full, over-capacity, past and duplicate bookings are rejected.
- ☐ A booking can be found with phone number + booking code.
- ☐ Cancelling returns the places to the event.
- ☐ The availability summary and event status are always correct.

15. Optional Bonus Features

Bonus features are **optional**. They are not needed for full core completion and they never replace a missing core requirement. Only start a bonus after the minimum deliverable is complete and working. Bonus features must not add a database, login or new major technologies.

- **Event sorting** — sort events by date, remaining places or popularity.
- **Category summary** — for each category, show the number of events and the total places booked.
- **Popular events** — a view of the events with the highest booking percentage.
- **Availability visualization** — a simple progress bar showing how full each event is.

16. Presentation Expectations

Your team will present the finished project **in English**. Your instructor will confirm the date and the time limit.

The presentation must include:

1. **The problem** — a brief explanation of who the product is for and what problem it solves.
2. **Product walkthrough** — a short tour of the main screens.
3. **Live demonstration** — the main user flows working live, including at least one invalid case being rejected with a clear message.
4. **Business rules** — an explanation, in your own words, of the most important rules your product enforces and how a user experiences them.
5. **Team** — every member speaks.

Before presenting, make sure the backend and frontend both start cleanly and the demo data is ready.

17. Evaluation Focus

You will be assessed on:

Area	What we look for
Functional completeness	The required features and screens exist and work.
Correct business behavior	The business rules are enforced exactly as described.
Frontend/backend integration	The frontend and backend genuinely work together as one working application.
User experience	Screens are clear, readable and easy to use without explanation.
Handling invalid cases	Invalid input and invalid actions are rejected with clear messages.
Team collaboration	Work is shared; every member contributed meaningfully.
Code organization and readability	Your code is understandable and reasonably organized.
Explaining your own work	Every member can explain the product and its rules.
Presentation quality	Clear English presentation with a working live demo.